

# VirtuBox Assessment: Software Tester (QA)

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## 1. Test Scenarios and Test Cases

I would test the application step by step. First I would check registration and login, and after that I would check the task related features. I would also check what happens when the user enters wrong, missing or unusual data.

### Registration

| Positive Cases                                                                                                                                                                                                                                                                                                                                                                                    | Negative Cases                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Edge Cases                                                                                                                                                                                                                                                                                                                                                 |
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| <ul style="list-style-type: none"><li>• I'd check that a user can register successfully when username, email and password are entered with valid details.</li><li>• Check that a valid email format is accepted (for example, <b>abc@gmail.com</b>, not <b>abc gmail.com</b>).</li><li>• Check that the registration completes successfully and the user can use the account for login.</li></ul> | <ul style="list-style-type: none"><li>• I'd check that an invalid email format is rejected (for example, <b>abc gmail.com</b>).</li><li>• Check that registration is not allowed when a required field such as username, email or password is empty.</li><li>• Check that the same email cannot be used to create another account, if email uniqueness is required.</li><li>• Check that invalid or weak password input is rejected according to the application's password rules, if such rules are defined.</li></ul> | <ul style="list-style-type: none"><li>• I'd check very long username, email or password input and make sure the application handles it properly.</li><li>• Check unusual but valid email characters and make sure valid input is not rejected unnecessarily.</li><li>• Check what happens when registration is submitted multiple times quickly.</li></ul> |

### Login

| Positive                                                                                                                                                                                                               | Negative                                                                                                                                                                                                                                                                                                                                                                                                            | Edge                                                                                                                                                                                                                     |
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| <ul style="list-style-type: none"><li>• I'd check that a registered user can log in using the correct email and password.</li><li>• Check that the user is taken to the task section after successful login.</li></ul> | <ul style="list-style-type: none"><li>• I'd Check that login fails when the password is incorrect.</li><li>• Check that the Forgot Password option works properly, <b>if provided</b>.</li><li>• Check that an invalid email format is rejected (for example, <b>abc gmail.com</b> instead of <b>abc@gmail.com</b>).</li><li>• Check that login is not allowed when the email or password field is empty.</li></ul> | <ul style="list-style-type: none"><li>• Check very long or unusual input in the email and password fields.</li><li>• Check repeated wrong-password attempts and make sure the application handles them safely.</li></ul> |

## Create Task

| Positive                                                                                                                                                                                                                                                                                                            | Negative                                                                                                                                                                                                                                                                                                                                                                                                 | Edge                                                                                                                                                                                                                     |
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| <ul style="list-style-type: none"><li>• I'd check that a task is created successfully when all required details are entered correctly.</li><li>• Check that a valid due date is accepted and the created task appears in the task list.</li><li>• Check that the task is saved correctly in the database.</li></ul> | <ul style="list-style-type: none"><li>• I'd check that a task cannot be created when required task details are missing.</li><li>• Check that invalid date input is rejected.</li><li>• Assuming past due dates are not allowed, check that a past date is rejected.</li><li>• Check that the application shows an error if the database save fails instead of showing a false success message.</li></ul> | <ul style="list-style-type: none"><li>• I'd check very long task details and make sure the application handles them properly.</li><li>• Check whether multiple quick clicks on Create produce duplicate tasks.</li></ul> |

## View Task List

| Positive                                                                                                                                                                                                                               | Negative                                                                                                                                                                                                      | Edge cases                                                                                                                                                                                                                                    |
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| <ul style="list-style-type: none"><li>• Check that all created tasks are displayed in the task list.</li><li>• Check that the correct task details are shown.</li><li>• Check that a newly created task appears in the list.</li></ul> | <ul style="list-style-type: none"><li>• I'd check that the application does not show incorrect or missing task information.</li><li>• Check that a deleted task no longer appears in the task list.</li></ul> | <ul style="list-style-type: none"><li>• I'd check the task list when there are no tasks and make sure the page does not break.</li><li>• Check the task list when there are many tasks and make sure the list still loads properly.</li></ul> |

## Edit Task

| Positive cases                                                                                                                                                                                                                                                    | Negative                                                                                                                                                                                                    | Edge                                                                                                                                                                                       |
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| <ul style="list-style-type: none"><li>• Check that an existing task can be edited successfully.</li><li>• Check that the updated task details are shown correctly after saving.</li><li>• Check that the updated values are also saved in the database.</li></ul> | <ul style="list-style-type: none"><li>• Check that invalid or missing required information is not saved while editing.</li><li>• Check that an invalid date, such as month 13 or 14, is rejected.</li></ul> | <ul style="list-style-type: none"><li>• Check saving without making any changes and make sure the task remains unchanged.</li><li>• Check editing a task with very long details.</li></ul> |

## Delete Task

| Positive                                                                                                                                                                                                                                                                  | Negative cases                                                                                                                                                                                                                 | Edge                                                                                                                                                                                                                                                                                        |
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| <ul style="list-style-type: none"><li>• Check that a task is deleted successfully after the user confirms deletion.</li><li>• Check that the deleted task no longer appears in the task list.</li><li>• Check that the database reflects the expected deletion.</li></ul> | <ul style="list-style-type: none"><li>• Check that cancelling the delete confirmation keeps the task unchanged.</li><li>• Check that an error is shown if deletion fails instead of showing a false success message.</li></ul> | <ul style="list-style-type: none"><li>• Check whether a confirmation message such as "Do you really want to delete this?" is shown before permanent deletion.</li><li>• Check multiple quick clicks on Delete and make sure the task is not processed incorrectly more than once.</li></ul> |

## 2. Input Validation & Error Handling

I would make sure that wrong or incomplete input does not break the application. The user should get a simple message explaining what went wrong. For example, an invalid email should show an email related error and an empty required field should be pointed out.

### **Positive Cases**

- Valid information should be accepted and the requested operation should complete successfully.
- After Create, Edit or Delete, the information shown in the application should match the expected database state.

### **Negative Cases**

- Empty required fields should show a clear validation message instead of allowing invalid data to be saved.
- Invalid email, password, date or other invalid input should be rejected with a simple and understandable error message.
- If there is a technical/database problem, the application should show a message such as "We are currently having a technical glitch, please try again later."
- The application should not show a success message when the requested operation actually fails.

### **Edge Cases**

- Very long input should be handled properly without breaking the page or application.
- If one feature has a technical problem, the rest of the application should continue working where possible instead of the whole application going down.
- User data and created tasks should have proper backup/recovery support so that important data is not easily lost.

## 3. Potential Bugs / Risk Areas

Since the application has not been executed, these are possible bugs or risk areas that I would look for. I would not call them confirmed bugs until they are actually tested.

### **1. Same email can be used for more than one account**

**Severity:** Major

**Reason / Impact:** A duplicate account can create confusion and may cause problems while logging in or identifying the correct user.

### **2. Valid registration fails**

**Severity:** Major

**Reason / Impact:** A user may provide correct details but still be unable to create an account, which prevents access to the application.

### **3. Newly created task is not saved in the database**

**Severity:** Critical

**Reason / Impact:** The user may see a success message, but the task can be lost. This affects the main purpose of the application.

### **4. Edit changes are not saved or old values are shown**

**Severity:** Major

**Reason / Impact:** The user may think the task was updated, while the old information remains in the application or database.

## 5. New task does not appear in the task list

**Severity:** Major

**Reason / Impact:** The task may have been created but the user cannot see it, making the task-management feature unreliable.

## 6. Deleted task still appears in the task list

**Severity:** Major

**Reason / Impact:** The UI may show information that should no longer be available, creating confusion about whether deletion actually happened.

## 7. UI and database show different task information

**Severity:** Major

**Reason / Impact:** Different data in the application and database can lead to incorrect information and loss of trust in the system.

## 8. Existing account is later treated as unregistered

**Severity:** Critical

**Reason / Impact:** A user could lose access to their account and previous tasks. This is a serious issue because it affects both login and stored user data.

## 9. Wrong login credentials are accepted

**Severity:** Critical

**Reason / Impact:** An incorrect password or other invalid credentials being accepted can allow unauthorized access to another user's account and tasks.

## 4. Overall Approach

My approach would be to first check the main features with correct information and make sure the normal flow is working. After that I would try wrong, missing and unusual inputs. I would also check whether the information remains correct when moving from one feature to another, for example Create to View, Edit to View and Delete to View. I would give more attention to login, account access, saving data and data consistency because these can affect the user's work.

**Note:** Some checks such as CAPTCHA, username uniqueness, password strength, past-date restrictions and permanent deletion are not specifically defined in the assessment. I would confirm the actual application requirements before treating these as definite failures